



HerShield — AI-Powered Women's Safety Platform

HerShield is an AI-powered women's safety platform that combines **Machine Learning**, **Real-Time Navigation**, **Anomaly Detection**, **Voice Recognition**, and **Emergency Response Automation** to provide intelligent safety assistance during travel.

Built using **Next.js**, **Node.js**, **Flask**, **PostgreSQL**, **Prisma**, **OpenStreetMap**, and **Machine Learning**.



Features



Safe Journey Navigation

- AI-based safest route recommendation
- Real-time route monitoring
- Live GPS tracking
- Weather-aware route prediction
- Distance & ETA calculation
- Interactive OpenStreetMap integration



AI Route Prediction

Instead of recommending the shortest path, HerShield recommends the **safest** route using machine learning.

Prediction considers:

- Crime score
- Lighting conditions
- Traffic density
- Hospital proximity
- Police station proximity
- Time of day
- Weather



Intelligent Emergency Detection

HerShield continuously monitors the user's journey. Emergencies can be triggered through:

- AI anomaly detection
- Scream detection
- Manual SOS button
- Failure to respond after an emergency prompt



AI Scream Detection

During navigation, microphone monitoring runs in the background.

Features:

- Continuous audio monitoring

- ML-based scream classification
- Confidence-based prediction
- Consecutive detection filtering
- Automatic emergency trigger

Live Location Sharing

When an emergency occurs, the system automatically shares:

- Live GPS location
- Vehicle number
- Emergency reason
- Google Maps location link

Email-Based SOS System

Instead of paid SMS services like Twilio, HerShield uses **Gmail SMTP** to instantly notify emergency contacts.

Emergency email contains:

- User's current location
- Vehicle number
- Emergency type
- Google Maps link
- Timestamp

Emergency Contacts

Users can securely save three emergency contacts during account setup. Each contact stores:

- Name
- Relationship
- Phone number
- Email address

Secure Authentication

- User registration
- Email OTP verification
- Secure login
- JWT authentication
- Password encryption using bcrypt
- Protected APIs

Real-Time Navigation

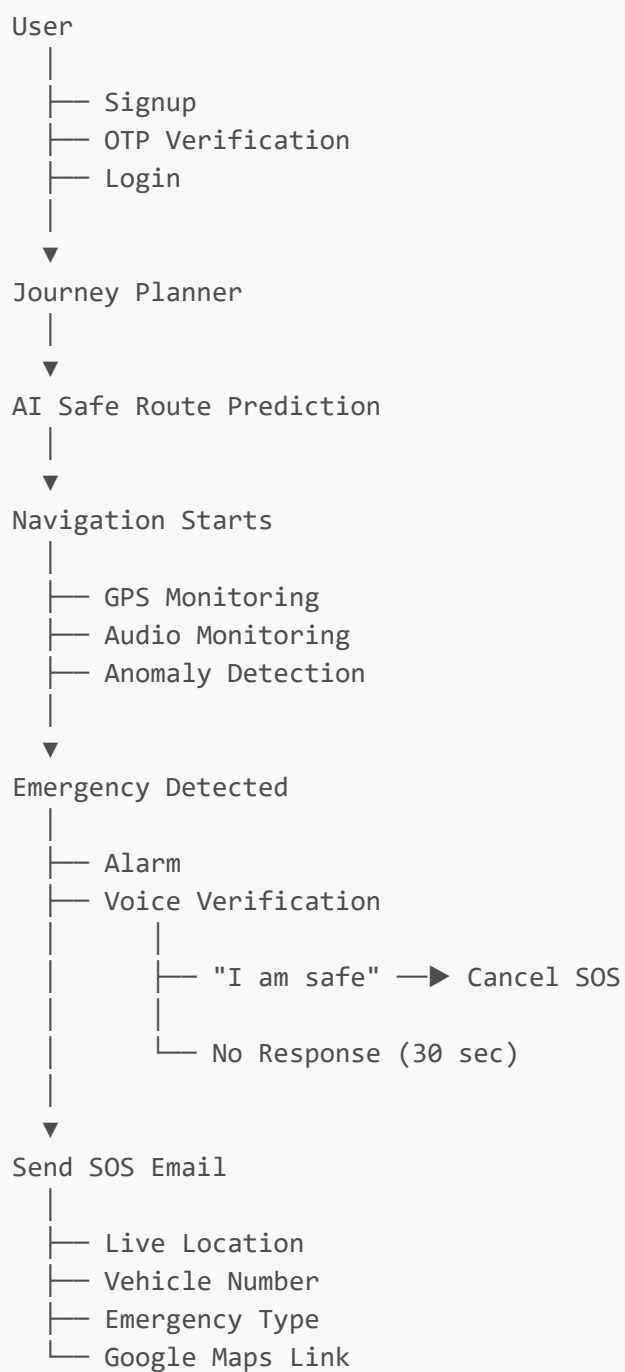
- Turn-by-turn navigation
- Live map updates
- Route visualization
- Journey status monitoring

Voice Verification

After an emergency is detected:

1. Alarm starts
 2. User gets **30 seconds** to respond
 3. User can say **"I am safe"**
-  If verified → SOS is cancelled
 -  If no response → Emergency notification is automatically sent

System Architecture



Tech Stack

Layer	Technologies
Frontend	Next.js, React, TypeScript, Tailwind CSS, Leaflet, OpenStreetMap
Backend	Node.js, Express.js, Prisma ORM, PostgreSQL, JWT Authentication, Nodemailer
Machine Learning	Python, Flask, Scikit-Learn, Random Forest, Isolation Forest, NumPy, Joblib
APIs Used	OpenStreetMap, Overpass API, Nominatim API, Gmail SMTP, Browser Speech Recognition API, Geolocation API

Project Structure

```
HerShield-Project
├── hershield-client
│   ├── app
│   ├── components
│   ├── lib
│   └── public
├── hershield-server
│   ├── controllers
│   ├── routes
│   ├── middleware
│   ├── prisma
│   └── uploads
├── hershield-ml
│   ├── api
│   ├── models
│   ├── training
│   └── utils
└── README.md
```

Installation

Clone Repository

```
git clone https://github.com/aan2136/hershield-project.git
```

Frontend

```
cd hershield-client
npm install
npm run dev
```

Runs on: <http://localhost:3000>

Backend

```
cd hershield-server npm install npm run dev
```

Runs on: <http://localhost:5000>

ML Service

```
cd hershield-ml pip install -r requirements.txt python app.py
```

 Runs on: <http://localhost:8000>

Environment Variables

Backend ([.env](#))

```
DATABASE_URL=your_postgresql_connection

JWT_SECRET=your_secret_key

SMTP_HOST=smtp.gmail.com
SMTP_PORT=587

SMTP_USER=your_email@gmail.com
SMTP_PASS=your_app_password
SMTP_FROM=your_email@gmail.com
```



User Workflow

Register ↓ Verify OTP ↓ Login ↓ Save Emergency Contacts ↓ Generate Safe Route ↓ Start Journey ↓ AI Monitoring Starts ↓ Emergency Detected ↓ 30-second Voice Verification ↓ SOS Email Sent



Future Enhancements

- ☐ Mobile application (Flutter / React Native)
- ☐ Hugging Face-based scream detection
- ☐ Offline emergency mode
- ☐ SMS & WhatsApp SOS integration
- ☐ Wearable device integration
- ☐ AI-based risk heatmaps
- ☐ Emergency audio recording
- ☐ Real-time police assistance
- ☐ Cloud deployment with Docker & Kubernetes



Contributors

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License

This project is developed for educational and research purposes.